Time-Series Momentum in Cryptocurrency:

A Published Edge That Has Aged — Regime-Dependence and Retail Feasibility (2016–2023)

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Abstract

Time-series momentum (TSMOM) — going long an asset whose own recent trailing return is positive and short (or flat) otherwise — is one of the more robust cross-asset findings in the momentum literature, documented across dozens of futures markets over multiple decades. Early cryptocurrency evidence (Liu and Tsyvinski, 2021) reported a strong Bitcoin TSMOM effect, but on a sample ending in May 2018, before the market structure a retail trader operates in today existed. This note asks two questions the series applies to every strategy: is the effect statistically real in the market regime that can actually be traded now, and — where the literature's numbers are positive — could a retail trader on a single exchange reach them? Reviewing the most recent, longest, and methodologically strongest crypto-specific test (Grobys et al., 2025; 30 large-cap coins, 2016–2023), the raw TSMOM effect is statistically indistinguishable from zero over the full sample (0.90%/week), marginally positive before July 2020 (1.74%/week, significant only at 10%), and negative and insignificant after July 2020 — the regime closest to today. The raw insignificance is driven by a single one-day 1,400% outlier; trimming it restores 1.51%/week (t = 2.63), underscoring heavy tails (power-law tail exponent below 3). A volatility-scaled overlay improves the full-sample figure (1.86%/week) but does not establish a base effect in the current regime for the overlay to improve. On the broader 2,500-coin value-weighted universe the factor is significantly negative (−3.40%/week) and is not reproducible on a single retail venue. Unlike the preceding notes in this series, this hypothesis is closed at the literature stage rather than with an own Binance replication, on the explicit ground that the strongest available study already reports no effect in the tradable regime. The contribution is a source-verified synthesis and a recency/regime-feasibility argument: a published edge that appears to have aged. This note does not claim trend-following "does not work"; it claims that this specific published evidence is not established as a currently tradable retail edge.

Keywords: cryptocurrency; time-series momentum; trend following; market regime; retail trading; replication discipline.

JEL classification: G11, G12, G14.

1. Why trend looked like the safe bet

Of the strategies this series has examined, trend-following — in its academic form, time-series momentum — arrived with the strongest pedigree. Moskowitz, Ooi, and Pedersen (2012) tested it across 58 liquid futures and forward contracts (equity indices, currencies, commodities, government bonds) over 1985–2009 and found the effect present in nearly every asset class, not concentrated in one corner of the data. The mechanism is simple and economically intuitive — an asset that has been rising tends to keep rising over the following weeks to months — and, importantly, the original construction already uses inversevolatility position sizing, so that no single instrument dominates portfolio risk. Volatility scaling is therefore part of the classical construction, not a later rescue invented for crypto (a point Section 6 returns to). Coming into this series, trend was the strategy expected to survive the crypto test most easily.

2. Two checks: is it real, and can retail reach it

Every note in this series applies two checks. The first is the one the academic evidence already addresses: is the effect statistically real, in a sample and methodology serious enough to trust. The second is one a published paper has no obligation to answer: even where the first check passes, can an individual with a retail-sized account, on a real exchange, using a realistic execution style, actually harvest the number reported — or does reaching it require a universe, a fee tier, or a moment in market history retail does not have? The two preceding notes each failed the second check differently — grid trading on an institutional maker-rebate fee tier (Part I), cross-sectional momentum on execution costs that consumed about 70% of an already-marginal spread (Part II). This note adds a third, distinct failure mode, developed below.

3. What time-series momentum measures

Time-series momentum differs from the cross-sectional momentum of Part II. Cross-sectional momentum ranks assets against each other and is long relative winners, short relative losers, regardless of the overall market. Time-series momentum instead looks at each asset's own trailing return in isolation: long if the asset's own price has risen over the lookback window, short (or, long-only, flat) if it has fallen. The comparison is an asset against its own past, not against its peers — a bet that recent direction persists, closer to what most retail traders mean by “the trend is your friend.”

4. The early crypto evidence — and its date

Liu and Tsyvinski (2021) found a strong Bitcoin TSMOM effect: a one-standard-deviation increase in a given week's return was associated with increases of 3.16%, 3.66%, 3.49%, and 1.50% in returns one, two, three, and four weeks ahead, reported as statistically significant. The detail that matters most here is the sample window: the Bitcoin, Ripple, and Ethereum samples end in May 2018 — a market with no deeply liquid, exchange-scale USDⓈ-M perpetual futures, minimal institutional participation, and none of the post-2020 structural changes later data must be tested against. A strong result on pre-2018 data is real evidence, but it is evidence about a market regime, not a permanent property of crypto — and, mechanically, from a period in which the retail execution infrastructure this series assumes (deep, broad Binance perpetuals, long and short, across dozens of large-cap alts) was in its infancy or did not yet exist.

5. The 2025 test: does the effect survive a longer, more recent sample?

Grobys, Kolari, Sandretto, Shahzad, and Äijö (2025) is the most recent and, on sample length and methodology, the strongest available crypto-specific test — 30 large-cap cryptocurrencies over 416 weekly observations, January 2016 to December 2023, spanning both the older pre-2018 regime and the post-2020 structure. As reported by the authors:

Specification (Grobys et al., 2025)	Result	Significance
Full sample 2016–2023, 30 coins, equal-weighted	0.90%/week	not significant
Pre-July 2020	1.74%/week	10% only
Post-July 2020 (regime closest to today)	negative	not significant
Full sample, single-outlier trimmed	1.51%/week	1% (t = 2.63)

Specification (Grobys et al., 2025)	Result	Significance
Volatility-scaled, 8-week window (full sample)	1.86%/week	5%
Broader 2,500-coin universe, value-weighted (full)	−3.40%/week	significant, negative

The contrast between the raw and outlier-trimmed 30-coin results is itself the finding. The authors trace the raw insignificance to a single episode: Mindol (MIN) rose from $0.54 to $7.86 on 23 December 2020 — roughly +1,400% in one day — which, in the short leg of a small portfolio, produced a single week's loss of about −255%. They estimate a power-law tail exponent below 3 (point estimates ≈ 2.45 for the plain strategy and 2.87 for the equal-weighted counterpart), implying variance that is not well-defined in the classical sense — a heavy-tailed regime closer to venture-style return distributions than to the roughly normal returns assumed by standard significance tests. The broader 2,500-coin value-weighted check is worse, not better: significantly negative and remaining negative (roughly −1.18% to −2.69%/week) across trimming approaches — the crash risk is not a small-basket quirk that diversification fixes.

6. Why volatility scaling does not change the verdict

Volatility scaling is part of the classical TSMOM construction, not an ad hoc modification introduced to rescue the cryptocurrency result. The relevant question is therefore not whether volatility scaling is legitimate, but whether the favorable full-sample figure survives the regime change identified in the same study. Grobys et al. report 1.86%/week for the volatility-managed strategy over the full 2016–2023 sample, while the post-July-2020 split reported for the raw strategy is negative and statistically insignificant; based on the results reviewed here, the study does not separately establish that the volatility-managed specification remains profitable and statistically significant in the post-July-2020 regime. The authors' own broader-universe check reinforces the caution: on the 2,500-coin value-weighted universe the raw factor is significantly negative, and after the same volatility scaling across three lookback windows average weekly payoffs remain −0.03%, −0.02%, and −0.03%, each significant at 1% — the crashes are removed, but the sign and significance of the underlying loss are not.

7. Closing the hypothesis without an own replication

7.1 The statistical ground, sharpened by regime. The series' working rule is to read the literature first and, where the evidence gives no reason to expect a different outcome on Binance data, close the hypothesis at the literature stage rather than reverse-engineer a net-of-costs backtest that the evidence already argues will fail its most basic hurdle — is the effect statistically real in the regime that matters. For cross-sectional momentum (Part II) the literature was split enough that running our own numbers was the only way to an answer. For time-series momentum, the strongest available study already finds no statistically significant evidence of the effect in the regime we would trade — the post-July-2020 subsample — with the closest positive number generated by a data window that predates today's market structure. On the tradable 30-coin universe this is a calendar problem, not a fee-tier or account-size one: no capital or execution skill changes which historical period the data comes from. The positive pre-July-2020 estimate (1.74%/week) belongs to a historical regime a trader today cannot revisit; the stronger trimmed (1.51%/week) and volatility-managed (1.86%/week) figures are full-sample estimates covering 2016–2023 and therefore do not, by themselves, establish that the effect remains present in the post-July-2020 subsample.

7.2 The retail-feasibility ground. The broader robustness universe — 2,500 coins, value-weighted, weekly-rebalanced — fails the second check on its own terms, independent of regime. It is an academic research construction across thousands of tickers, not a directly reproducible single-venue retail portfolio: no exchange, then or now, has listed anything close to 2,500 coins with tradable perpetual depth on both sides simultaneously (Binance's futures product lists on the order of a few hundred pairs, and did not exist before September 2019, covering more than half the sample). A retail trader cannot open the short leg — often either leg — for most of that universe on a single venue. And the access problem is compounded by the numbers: that universe is significantly negative and stays negative under every risk-managed and trimming approach. There is no “if only I could access it” that rescues this case.

Conducting an independent 30-coin replication would re-derive, at real cost, a result the literature already establishes with more data than is practical to reproduce at this project's scale — in a regime where the same authors' split-sample test already reports no statistically significant effect to find. That is the same discipline that led this series to publish two closed hypotheses rather than force a third variant “to save it.” A closed hypothesis is still a result.

8. Verdict

Time-series momentum has genuine academic standing on traditional assets (Moskowitz, Ooi, Pedersen), and nothing here disputes the mechanism as tested there. The crypto-specific evidence is regime-dependent: strong evidence on pre-2018 data (Liu and Tsyvinski), followed by a longer, more recent test (Grobys, Kolari, Sandretto, Shahzad, and Äijö) that finds the effect statistically indistinguishable from zero in the post-July-2020 regime — and negative in point estimate. The raw full-sample result is highly sensitive to extreme, fat-tailed episodes: removing a single extraordinary short-leg event changes the estimate from statistically insignificant to significant. A volatility-managed specification improves the full-sample number, but the study does not separately establish that it survives the regime change identified within it. The hypothesis closes for two reasons: statistically, on the one tradable universe (30 large-cap coins), the effect is not established in the regime that matters; and on the broader universe the literature also tested (2,500 coins), the published number was never a reproducible single-venue retail portfolio and is significantly negative regardless. This is a published edge that appears to have aged — real, or close to real, in an earlier window, and neither established nor reachable in the current one. That is a narrower claim than “trend-following does not work.”

In short: the historical evidence for cryptocurrency time-series momentum is substantial, but the evidence reviewed here does not establish that the effect persists as a statistically reliable, directly tradable retail edge in the more recent regime. **Status: closed at the literature gate — not disproven, not replicated.**

9. Implications for a retail trader

If you are considering a simple time-series trend system on crypto today — long when recent price action is up, flat or short when down — the evidence summarized here indicates that the most recent and most careful academic test of exactly this idea, on a multi-year sample that includes the current market structure, does not find a statistically reliable edge in the period that matters. The positive-looking results come from an older regime and from a few extreme episodes, not from a broad, dependable drift one could expect to harvest systematically going forward.

The failure mode differs in kind from the previous notes. Grid's problem (Part I) was a fee tier — fixable in principle with the right account. Cross-sectional momentum's problem (Part II) was costs eating a real but marginal edge — fixable in principle with lower fees or a larger edge. Trend's problem, on the one universe accessible on Binance today, is fixable by none of these: the evidence that looks positive belongs to a period of market history that has already happened. Neither greater account size nor better execution can resolve this temporal limitation. This does not mean every systematic trend approach is worthless — this note tests one specific published formulation of time-series momentum, not every way a trader might define “trend.” It means that citing academic trend-following's traditional-asset record, by itself, is not sufficient evidence that the same mechanism is currently a tradable edge on crypto.

References

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Companion series (GitHub): Searching for Edge — github.com/Silent47boryara/searching-for-edge. Part I: The Grid Bot Illusion. Part II: Momentum — Evidence from a Binance Replication. This note is Part III (literature-based synthesis; no own replication).